# Cody Slater

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## About

MD student and PhD candidate. I build models that predict and design RNA modifications, and the assays and in vivo tools that test those designs in cells and the mouse brain.

## Education

**Doctor of Medicine** — New York, NY
Columbia University — Expected 2027

**Doctor of Philosophy, Biomedical Engineering** — New York, NY
Columbia University — Expected 2027

**Master of Philosophy, Biomedical Engineering** — New York, NY
Columbia University — Expected 2026

**Master of Science, Biomedical Engineering** — New York, NY
Columbia University — May 2023

**Bachelor of Science, Aerospace Engineering** — Worcester, MA
Worcester Polytechnic Institute — May 2016

## Research & Teaching Experience

**Vicković Technology Innovation Lab, New York Genome Center**, Graduate Researcher (Ph.D.) — New York, NY
*deep learning · PyTorch · GPU HPC · single-cell & spatial transcriptomics · lentivirus · AAV · Cas9 · Cas13* — 2023 – present

- Leading development of large specialized models for RNA modification prediction and generative design.
- Established the first selection-based functional assay for RNA modifications; 10,000+ construct pools.
- Engineering AAV vectors for cell-type-specific perturbation in mouse brain, from prep to tissue processing.
- Building a spatial transcriptomics toolset to model in vivo cell-type-specific genetic perturbations.

**Department of Neurological Surgery, Columbia University**, Graduate Researcher — New York, NY
*intraoperative microneurography · DBS electrodes · human trigeminal ganglion · OR hardware design* — 2024 – present

- Built the intraoperative recording setup for human trigeminal ganglion, using DBS electrodes.

**Trends in Genomics, Columbia University**, Teaching Assistant — New York, NY

- Co-designed the course for its first offering (2025). — Spring 2025 & 2026

**Neural Engineering and Control Lab, Columbia University**, Graduate Researcher (M.S.) — New York, NY
*Neuropixels · GRAB sensors · 2-photon · optogenetics · chemogenetics · stereotaxic surgery · histology · behavior* — 2018 – 2023

- Co-created neurOS, a Simulink Real-Time platform for multimodal closed-loop experiment control.
- Designed and implanted brain interfaces for Neuropixels, fiber photometry and 2-photon imaging.
- Developed the lab's in vivo perturbation assay and its cell-type-specific transcriptomic readout.
- Launched a longitudinal locus coeruleus stimulation study in an Alzheimer's model; basis of an R01 line.

**Department of Biomedical Informatics, Columbia University**, Student Researcher — New York, NY
*retrospective case series · clinical data analysis* — 2020

- Co-led 30 medical students and 18 clinicians characterizing the first COVID-19 wave in New York.

## Professional History

**MIT Lincoln Laboratory / MIT Haystack Observatory**, Engineer, Advanced Sensors and Techniques Group — Westford, MA
*radar systems · RF hardware · thermal design · field deployment · computer vision* — 2016 – 2018

- Designed the structure and cooling for a solid-state radar transmitter, fielded in the Marshall Islands.
- Built a stand-alone traveling-wave-tube test facility; led the Millstone 84-foot azimuth encoder upgrade.
- Led a team adapting computer vision for autonomous UAV flight through an obstacle course.

**MIT Lincoln Laboratory**, Student Researcher, Systems Engineering Group — Lexington, MA
*systems design · spacecraft propulsion · vacuum testing* — 2015 – 2016

- Designed, built and characterized an experimental pico-satellite monopropellant thruster in vacuum.

**GE Aviation / Unison Industries**, Aerospace Engineering Intern — Jacksonville, FL
*manufacturing engineering · design automation* — 2014

- Automated the build instructions for jet-engine cable harnesses, generated from the design files.

## Publications & Papers

**Intraoperative microneurographic recording from the human trigeminal ganglion using standard deep brain stimulation electrodes**
**C. Slater**, …, R. F. Sekula Jr.
Submitted (2026)

**Intracranial delivery of neuron-preferential lipid nanoparticles for gene-editing activity in mouse brain**
S. S. Cai, **C. Slater**†, …, K. W. Leong
Cell Reports Physical Science 7(9), 103514 (2026) [DOI] *† equal contribution*

**Behavioral state-dependent norepinephrine dynamics in the primary somatosensory and prefrontal cortices during tactile detection tasks**
C. Kelley, T. Lantin, **C. Slater**, …, Q. Wang
bioRxiv preprint (2026) [bioRxiv]

**Alpha modulation of spiking activity across multiple brain regions in mice performing a tactile selective detection task**
C. Kelley, **C. Slater**, …, Q. Wang
European Journal of Neuroscience 62(12), e70356 (2025) [DOI]

**Bidirectional modulation of somatostatin-expressing interneurons in the basolateral amygdala reduces neuropathic pain perception in mice**
A. Apte, J. Fernald, **C. Slater**, …, Q. Wang
Frontiers in Pain Research 6, 1602036 (2025) [DOI]

**Brain-wide cell-type-specific noradrenergic modulation of the transcriptome**
**C. Slater**, …, Q. Wang
bioRxiv preprint (2025) [bioRxiv]

**The Neuromodulatory Role of the Noradrenergic and Cholinergic Systems and Their Interplay in Cognitive Functions: A Focused Review**
**C. Slater**, …, Q. Wang
Brain Sciences 12(7), 890 (2022) [DOI]

**Alzheimer's disease: An evolving understanding of noradrenergic involvement and the promising future of electroceutical therapies**
**C. Slater**, Q. Wang
Clinical and Translational Medicine 11(4), e397 (2021) [DOI]

**Characterising the long-term clinical outcomes of 1190 hospitalised patients with COVID-19 in New York City: a retrospective case series**
S. M. Shoucri, …, **C. L. Slater**, …, J. E. Zucker
BMJ Open 11(6), e049488 (2021) [DOI]

**Characterization and clinical course of 1000 patients with coronavirus disease 2019 in New York: retrospective case series**
M. G. Argenziano, S. L. Bruce, **C. L. Slater**†, …, R. Chen
BMJ 369, m1996 (2020) [DOI] *† equal contribution*

**Interventions to reduce short-wavelength ("blue") light exposure at night and their effects on sleep: A systematic review and meta-analysis**
A. Shechter, K. A. Quispe, J. S. Mizhquiri Barbecho, **C. Slater**, L. Falzon
Sleep Advances 1(1), zpaa002 (2020) [DOI]

**An Interprofessional Approach to Clinical Workflow Evaluation Focused on the Electronic Health Record Using Time Motion Study Methods**
J. Schwartz, J. Elias, **C. Slater**, K. Cato, S. C. Rossetti
AMIA Annual Symposium Proceedings, 1187–1196 (2019)

## Fellowships & Grants

**Van C. Mow Fellowship** — Sep 2021

**National Institute on Aging T35AG044303** — Mar 2019
Brief Research in Aging and Interdisciplinary Neurosciences (BRAIN) Program

## Presentations

**Grounding RNA prediction models in structure and function** — New York, NY, Oct 2026
Irving Institute for Cancer Dynamics Research Meeting, Columbia University. Speaker.

**Whole-brain, cell-type-specific transcriptomic profiling of in vivo neurochemical perturbation** — Chicago, IL, Oct 2024
Society for Neuroscience. Presenting author.

**Leveraging large-scale electrophysiology array recording to identify neural biomarkers for improved closed-loop neuromodulation for chronic pain** — Chicago, IL, Oct 2024
Society for Neuroscience. Contributing author.

**Noradrenergic modulation of population activity in the primary somatosensory cortex, striatum, and amygdala in a tactile selective detection task** — Chicago, IL, Oct 2024
Society for Neuroscience. Contributing author.

**Locus coeruleus modulation of population activity in the awake somatosensory cortex** — Washington, DC
Society for Neuroscience. Contributing author. — Nov 2023

**Lipid nanoparticle-mediated gene editing in the mouse brain via intracranial delivery** — Seattle, WA
Biomedical Engineering Society. Contributing author. — Oct 2023

**Noradrenergic dynamics in the sensory and prefrontal cortices during a perceptual decision-making task** — San Diego, CA
Society for Neuroscience. Presenting author. — Nov 2022

**Functional noradrenergic regulation of the transcriptome during decision making under varying uncertainty** — New York, NY
Center of Excellence in the Neuroscience of Decision-Making Kickoff, Columbia University. Presenting author. — Oct 2022

**Development of a neural network architecture for optimal prediction of EEG alpha wave power** — New York, NY
Northeast Bioengineering Conference, Columbia University. Presenting author. — Apr 2022

**Targeted neural stimulation to improve neuropathic pain treatment** — New York, NY
Northeast Bioengineering Conference, Columbia University. Presenting author. — Apr 2022

**MIDA-based finite element modeling to optimize electrotherapy configurations** — New York, NY
Northeast Bioengineering Conference, Columbia University. Contributing author. — Apr 2022

**Long-term locus coeruleus stimulation reduces Aβ pathology in a mouse model of neurodegeneration** — New York, NY
Medical School Research Symposium, Columbia University. Presenting author. — May 2021

**Synthetic neuromodulation as a potential treatment for Alzheimer's disease** — New York, NY
NIH Summer Research Symposium, Columbia University. Speaker. — Aug 2019

**The temporal expression of consciousness** — Worcester, MA
Class of 1879 Award Ceremony and Reception, WPI. Speaker. — Mar 2017

## Leadership

**President, Columbia Health Tech Assembly** — Jun 2020 – Aug 2022

**Senior Advisor, Innovative Medicine Club** — Aug 2020 – Aug 2022

**Student Member, VP&S Curriculum & Education Discover Subcommittee** — 2020 – 2021

**Ethics Representative, Medical School Class Council** — Aug 2018 – Jul 2021

**Treasurer, WPI Chapter of AIAA** — Jan 2014 – Jan 2015

## Honors & Awards

**Best Session Poster, Medical School Research Symposium, Columbia University** — May 2021

**Roy and Diana Vagelos Scholarship** — 2018 – 2021

**D. & D. Merksamer Scholarship** — 2018 – 2021

**WPI Class of 1879 Prize for Outstanding Original Work in the Humanities and Arts** — Mar 2017

**WPI Graduation with Honors and High Distinction** — May 2016

**WPI Presidential Scholarship** — 2012 – 2016

**Most Impactful Project Award, WPI Interactive Qualifying Project** — May 2015